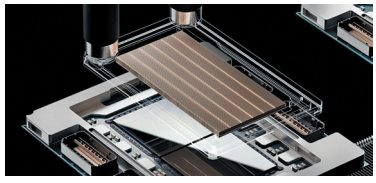
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Next-generation dual-architecture processor

IBM has announced a 2nm dual-architecture mainframe processor for future IBM Z and LinuxONE systems that executes both Arm and IBM Z/LinuxONE instructions concurrently



on each core. Operating at 5.7GHz, the 11-core design integrates an on-chip DPU for I/O acceleration and AI inference accelerators. The platform natively brings IBM's enterprise-grade capabilities to Arm workloads, including hardware-level fault detection and recovery, encryption, secure key management, and AI acceleration.
IBM
https://www.ibm.com/in-en

6nm SoCs for edge AI applications

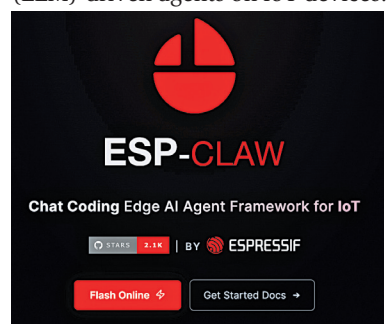
Amlogic has announced two 6nm systems-on-chips (SoCs), C305X2 and A123X, for edge AI applications. Both use the Arm Cortex-A320 CPU



based on the Armv9 architecture with SVE2 vector extensions, yielding a 34 per cent SPECint2006 performance increase over Cortex-A35 and up to 10 × machine-learning processing uplift. The chips integrate a neural hardware SED engine and an 8 TOPS transformer-native ADLA3 NPU for local ViT and compact LLM execution. The flagship A123X adds a secondary 4 TOPS ADLA2 NPU for dedicated object tracking. Both SoCs target battery-powered IP cameras, industrial machine vision, smart home cameras, and IoT devices.
Amlogic
https://www.amlogic.com

Edge AI agent framework for IoT

Inspired by the agent-runtime model and adapted for ESP32 series microcontrollers, ESP-Claw is a framework for running large language model (LLM)-driven agents on IoT devices. It



supports event-driven responses, context retention, and local actions while integrating with sensors, device states, and physical-world inputs. Developers can build local or hybrid AI agents that process inputs, make decisions, and control connected devices. The framework runs in resource-constrained edge environments and seamlessly connects to cloud-based LLMs when required, targeting smart home, industrial, and consumer AIoT systems.
Espressif Systems
https://www.espressif.com

Chip stacking logic units in multiple layers

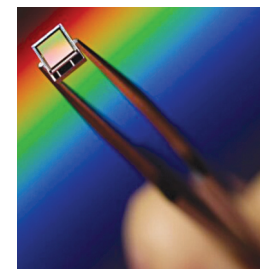
Huawei has introduced the Kirin 9050 Pro chip, which combines a LinxiCore CPU, Maleoon GPU, and Huawei Da Vinci Architecture NPU for computing, graphics, and on-device AI workloads. The company claimed that device performance improves by 42 per cent compared to the previous generation through coordination across hardware, software, chip, and cloud systems. The



Kirin 9050 Pro is described as the first high-performance processor to use LogicFolding technology, which stacks logic units in multiple layers within a single chip instead of keeping them on a single plane. This reduces the distance signals travel between logic units, which is intended to lower signal transmission latency and boost overall performance.
Huawei
https://www.huawei.com/en

SPAD sensor integrating multiple photon statistics processing on a chip

Singular Photonics has launched Litavis, a single-photon avalanche diode (SPAD) image sensor that integrates imaging,



timing, histogramming, and photon statistics processing on a single chip. Its software-configurable sensing architecture combines photon-counting imaging, programmable time gating, and photon timing within a SPAD array. The sensor delivers continuous 256 × 256-pixel photon-counting imaging in parallel with a 64 × 64 macropixel grid providing picosecond-resolution photon timing. It operates in intensity, timing, and histogramming modes simultaneously, enabling software-driven parameter adjustments without hardware redesigns. On-chip processing of photon